

Skill-20

Rollno:2520030011

Program:

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tejaswini@teju: ~/OS_Lab/skil × + v
GNU nano 8.7.1
// OSSP Skill 20 - SIGTSTP and Process Groups
// Demonstrates process groups, SIGTSTP suspension,
// SIGCONT resume, and group-based signal delivery.

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>

int main(void) {
    printf("== Skill 20: SIGTSTP + Process Groups ==\n");

    pid_t pid = fork();
    if (pid == -1) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        setpgid(0, 0);

        printf("Child pid=%d pgid=%d\n", getpid(), getpgrp());
        fflush(stdout);

        for (int i = 1; i <= 8; i++) {
            printf("Child running: %d\n", i);
            fflush(stdout);
            sleep(1);
        }

        _exit(0);
    }
}
```

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    sleep(2);

    printf("Parent sends SIGTSTP to the child process group.\n");
    if (kill(-pid, SIGTSTP) == -1)
        perror("SIGTSTP");

    int status;
    waitpid(pid, &status, WUNTRACED);

    if (WIFSTOPPED(status))
        printf("Child stopped by signal %d.\n", WSTOPSIG(status));

    printf("Parent resumes the group using SIGCONT.\n");
    if (kill(-pid, SIGCONT) == -1)
        perror("SIGCONT");

    waitpid(pid, &status, 0);

    printf("Child completed after resume.\n");
    return 0;
}

```

Output:

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tejaswini@teju:~/OS_Lab/skill$ nano skill_20.c.c
tejaswini@teju:~/OS_Lab/skill$ gcc skill_20.c.c -o skill_20
tejaswini@teju:~/OS_Lab/skill$ ./skill_20
=== Skill 20: SIGTSTP + Process Groups ===
Child pid=1486 pgid=1486
Child running: 1
Child running: 2
Parent sends SIGTSTP to the child process group.
Child stopped by signal 20.
Parent resumes the group using SIGCONT.
Child running: 3
Child running: 4
Child running: 5
Child running: 6
Child running: 7
Child running: 8
Child completed after resume.
tejaswini@teju:~/OS_Lab/skill$ |

```